

Customer Analysis

1. Project Overview

Purpose

- This end-to-end customer analysis project examines shopping behavior using a transactional dataset containing 3,900 customer purchases across multiple product categories.
- The primary objective is to convert raw customer transaction data into meaningful business insights that support strategic decision-making.
- The analysis focuses on customer demographics, purchasing habits, seasonal buying behavior, subscription patterns, and product preferences.

Business Value

- Helps identify customer trends and spending behavior.
- Supports targeted marketing campaigns and customer retention strategies.
- Provides a complete analytics pipeline using Python, PostgreSQL, and Power BI.

2. Dataset Summary

Dataset Information

- Contains 3,900 rows and 18 columns.
- Includes customer demographics such as Age, Gender, Location, and Subscription Status.
- Captures purchase details including Item Purchased, Category, Purchase Amount, Season, Size, and Color.
- Behavioral attributes include Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, and Shipping Type.

Data Quality

- Detected 37 missing values in the Review Rating column.
- The dataset is suitable for customer segmentation, revenue analysis, purchase trend analysis, and behavioral analytics.

3. Exploratory Data Analysis using Python

Data Preparation

- Imported the dataset using pandas.
- Used `df.info()`, `df.describe()`, and null-value analysis to understand the dataset.
- Handled missing review ratings by replacing them with the median rating within each product category.

Feature Engineering

- Standardized all column names using `snake_case`.
- Created `age_group` for demographic analysis.
- Created `purchase_frequency_days` for behavioral analysis.

- Validated data consistency and removed the redundant promo_code_used column.
- Loaded the cleaned dataset into PostgreSQL for SQL analysis.

4. Data Analysis using SQL

Business Questions Solved

- Compared revenue generated by male and female customers.
- Identified high-spending customers who used discounts.
- Ranked the top five highest-rated products.
- Compared average purchase amounts across shipping methods.
- Evaluated subscribers versus non-subscribers.
- Found products with the highest percentage of discounted purchases.
- Segmented customers into New, Returning, and Loyal groups.
- Listed the top three products in every category.
- Analyzed repeat buyers and subscription behavior.
- Calculated revenue contribution by different age groups.

Outcome

- These SQL queries transformed raw data into business-ready insights for decision-making.
- They highlighted purchasing patterns, customer value, and opportunities for targeted marketing.

5. Business Recommendations

Recommendations

- Promote subscription plans through exclusive offers and personalized benefits.
- Strengthen loyalty programs to improve customer retention and repeat purchases.
- Optimize discount strategies to balance customer acquisition with profitability.
- Prioritize high-rated and frequently purchased products in marketing campaigns.
- Focus promotional activities on customer segments and age groups contributing the highest revenue.
- Continuously monitor customer behavior using dashboards for data-driven decisions.